

Note 15.1 Double and Iterated Integrals over Rectangles

Double Integrals

Double integrals are **definite integrals**, and functions with more variables have **no infinite integrals**.

The double integrals can be obtained by Riemann sum of small rectangles. Denoted by

$$\iint_R f(x, y) dA, \iint_R f(x, y) dx dy$$

on the region R .

We can see the function as the **height** of each rectangular box and the integral is the sum of the **volume**. Similar to **single integral**.

We can also see the function as the **density** of unit area and the integral is the **mass** of the piece. Similar to **triple integrals**.

Fubini's Theorem for Calculating Double Integrals

Theorem 1--Fubini's Theorem(First Form)

For a continuous function $f(x, y)$ through the rectangular region $R : a \leq x \leq b, c \leq y \leq d$,

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$

It is to integrate the small **area slice** through an axis first, then integrate **volume** through another axis.

And it is called **iterated or repeated integral**.

When integrating inner function, see another variable as a **constant**.

Exercises